

NATIONAL JUNIOR COLLEGE

Higher 2



CANDIDATE
NAME

BIOLOGY
CLASS

REGISTRATION NO.

BIOLOGY

9747/03

JC2 Preliminary Examinations

8 Sept 2008

Paper 3

1 Hour 30 min

READ THESE INSTRUCTIONS FIRST

Write your name and registration number (e.g. 05IP0409 or 07S0620) and Biology Class (e.g. 2bi2E) clearly on the cover page.

Write your Name, Registration No. and Biology Class on the writing paper that you write the essay on.

Write in dark blue or black pen.

Answer **all** questions.

At the end of the exam, staple your essay separately and hand in question paper and essay separately .

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINERS' USE ONLY	
1	
2	
3	
4	
TOTAL (/65)	

This paper consists of **10** printed pages including cover page.

SECTION A: 45 MARKS

Answer all questions.

Question 1

You are studying a specific gene named *LC1* in mice, and you want to express that mouse gene in *E. coli*. To do that you must insert the mouse gene into the bacterial plasmid using restriction enzymes.

These are the restriction enzymes you can use. Their respective restriction sites are indicated as **X**: for *Xba*I, **N**: for *Nde*I, **S**: for *Sal*I, **E**: for *Eco*RI and **K**: for *Kpn*I.

Below is a map of the area of the mouse genome surrounding *LC1* in which you are interested.

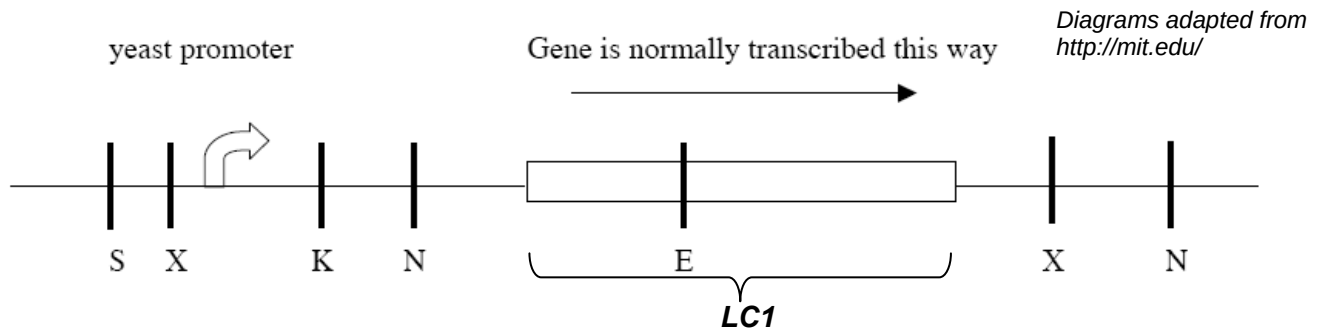


Fig. 1.1 Map of *LC1* mouse gene and adjacent regions. Curved arrow indicates the position of the *LC1* mouse gene promoter.

This is the map of the cloning vector which has an ampicillin resistance gene somewhere distant from the restriction sites.

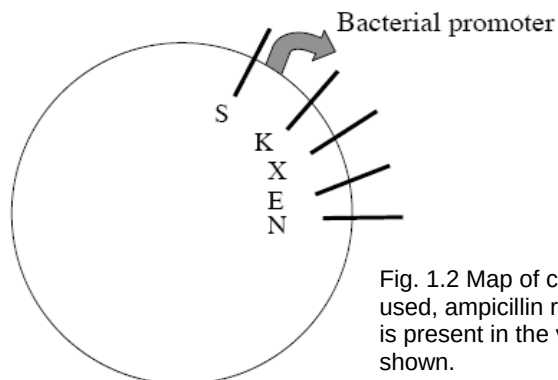


Fig. 1.2 Map of cloning vector used, ampicillin resistance gene is present in the vector but not shown.

(a) Besides the regions in the cloning vector already mentioned, state one other region in the vector that is required and why it is necessary. [2]

- **Origin of replication [1]**
- **Sequence to which DNA polymerase (and helicases) will bind**
- **to initiate replication of the vector with the gene insert / recombinant vector OR so that plasmid can replicate independently**

ALTERNATIVE Possible ANS

- **Another (or named – not Ampicillin) antibiotic resistance gene / β -galactosidase gene within the region where the restriction sites are [1]**
- **So that the insertion of *LC1* at the restriction site will disrupt this antibiotic resistance gene / β -galactosidase gene**
- **And allow for selection of bacteria with the recombinant plasmid.**

(b) i) After much consideration, NdeI was used to cut both the gene of interest and the vector. Explain why NdeI was chosen to cut the gene of interest and the vector. [2]

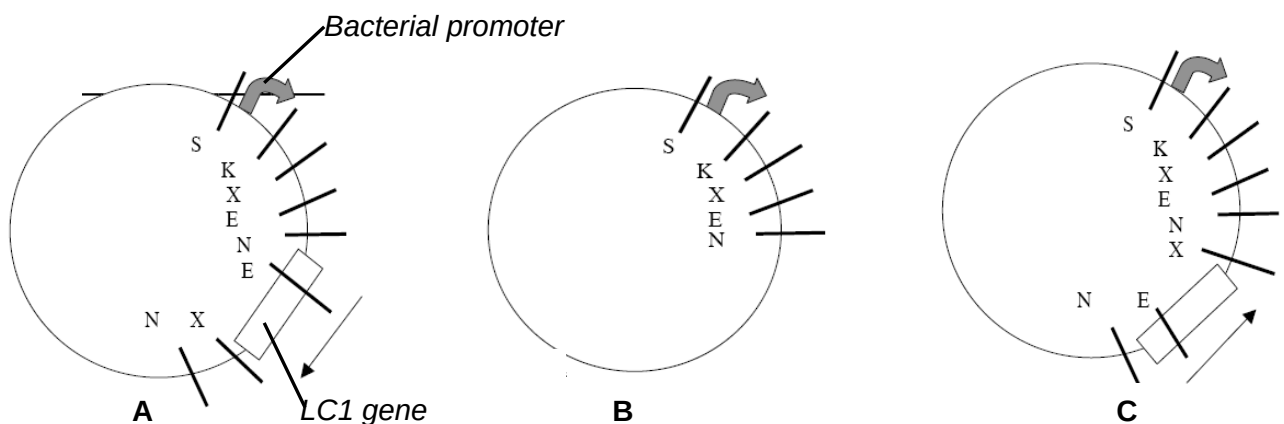
- **There are two NdeI restriction sites flanking / each on either side of the gene of interest**
- **and so will cut out the whole gene of interest from the surrounding genome**
- **Also cutting at the two NdeI restriction sites will exclude the mouse promoter,**
- **good because the bacterial promoter is the one that is required to initiate transcription in the bacterial cell / mouse promoter not needed to initiate transcription, cannot initiate transcription.**

ii) Explain what occurs from this point mentioned in part (b)i) , in order to obtain bacteria that contain the recombinant plasmid. (NdeI produces sticky ends) [3]

- **Complementary sticky ends of gene of interest and vector will anneal together**
- **by hydrogen bonding**
- **Ligase**
- **will be used to seal the nicks / close the gaps between the gene of interest and the vector**
- **by catalyzing the formation of phosphodiester bonds**
- **Recombinant plasmid will be introduced into bacterial cells / transformation**
- **via CaCl₂ treatment followed by heat shock OR electroporation**
- **Screening for bacteria with recombinant plasmid mentioned**

c) Three possible ligated DNA products that may result from the ligation which will confer ampicillin to the transformed bacteria are shown below.

Which of the three ligated DNA products is the most likely to allow for successful production of the mouse protein from the bacterial cells? Explain. [3]



- **A is the one you want. [1]**
- **You want the bacterial promoter to drive transcription of the mouse gene in the correct orientation,**
- **so that the correct strand of DNA is used as a template in transcription.**
- **B is the vector re-ligated to itself and does not have the gene inserted.**
- **C would result in the wrong strand of DNA being used as the template for transcription.**

(d) Through many months of hard work you discover that this gene LC1, when mutated, decreases the incidence of cancer in mice. You now know the sequence of this gene, and you really want to see if there is a similar gene in humans. However to date, no such gene has been found in humans. So, you set out to find the human homolog of LC1 by screening a human library.

i) Assuming you could only choose to screen one human library, what kind of human library would you use? Explain your answer briefly. [2]

- ***Because the LC1 gene may only be expressed in certain tissues of mice, but no information was given as to which are the tissues that express the gene,***
- ***you may choose to screen a genomic library [1]***
- ***so that all the genes present in a human can be screened***

ii) How would you screen the human library for the homolog of LC1? [3]

- ***Because the sequence of the LC1 gene from mice is known,***
- ***you could make a probe complementary to part of the mouse sequence.***
- ***A human homolog, if it exists, would have significant sequence similarity / partially, large percentage complementary to probe.***

Reject: Identical sequence / will be complementary. [1]

- ***The hybridization could be performed at a temperature that would allow probe binding to similar but not necessarily identical sequence. [1]***

Question 2

In recent years, one of the most exciting areas of research is in stem cells. The biology of mouse stem cells has been studied in detail for many years. This led to the development of techniques to isolate stem cells from human embryos and the ability to grow these cells in the laboratory. This breakthrough has made the vision of regenerative or reparative medicine a distinct possibility in the near future.

(a) i) What do you understand by the term regenerative medicine? [1]

Treatment in which stem cells are induced to differentiate into the specific cell type required to repair damaged or destroyed cells or tissues.

ii) Explain three unique features of stem cells. [3]

Unspecialized. They do not perform any tissue specific functions.

Self-renewing. have the ability to divide indefinitely into the same/ identical non-specialized cell types – often throughout the life of the organism.

have the potential to give rise to specialized cells, / able to differentiate, under the correct physiological or experimental condition,

or stem cells can be induced to differentiate into specific tissue lineages (i.e. nerve cells of the nervous system).

When the mouse stem cells are not isolated for research, they continue with their normal functions in the living organism.

iii) Using appropriate examples, explain what these normal functions are. [4]

- **Early embryonic cells (totipotent) formed just after the egg has been fertilized**
- **have the potential to develop into any specialized cells in the living organism.**
- **divide and multiply into a hollow ball - the blastocyst,**
- **consisting only of an outer cell mass which in a pregnancy would later form the placenta, and an inner cell mass, which would become the fetus**
- **the embryonic stem cells (the inner mass of the blastocyst) can develop into any of the hundreds of cell types in the adult body of living organism (except the extra-embryonic membranes.)**
- **these pluripotent stem cells continue to divide, they begin to specialize further.**
- **become adult stem cells (multipotent) which can give rise to several other cell types within tissues / organs.**
- **Adult stem cells typically generate the cell types of the tissue in which they reside.(for wear and tear / growth / development of organism)**
- **hematopoietic stem cells / adult stem cells (in bone marrow) – normally give rise to the many types of blood cells(red blood cells, white blood cells and platelets) in normal living organism.**

(b) Today, many expectant mothers and couples are asking questions such as "Should I bank my baby's cord blood?" Give three advantages of banking baby's cord blood for future use. [3]

- **After birth, blood from the umbilical cord contains stem cells which are multipotent, i.e. they can differentiate into several cell types.**
- **Hematopoietic stem cells may differentiate(under controlled conditions) into: three major types of brain cells, skeletal muscle cells, cardiac muscle cells, and liver cells.**
- **Cord blood can be used to treat patients who have undergone chemotherapy to destroy their bone marrow due to cancer or other blood-related disorders.**
- **These stem cells are genetically identical to the baby and will not trigger immune response when introduced into the same adult in future.**
- **Increase chances of a match should his siblings / parents require stem cell therapy.**
- **It also provides a safe, painless and non-controversial means of acquiring stem cells.**
- **Is generally preferred in stem cell transplantation because of a lower complication rate compared to bone marrow transplantation**
- **Can be readily made available should the need arise, thus eliminating the time needed to search for a matching donor's stem cells and allowing treatment to begin promptly**

Gene therapy is a rapidly growing field of medicine in which genes are introduced into the body to treat diseases. However, there are still many social and ethical considerations for the use of gene therapy.

(c) Discuss four of these considerations. [4]

- **People began to wonder whether it is possible to distinguish between "good" and "bad" uses of the gene modification techniques, and whether the potential for harmful abuse of the technology should keep us from developing more techniques.**
- **There is the question of tempering with genes – against nature / playing God.**
- **In SCNT, embryo destroyed to accommodate a nucleus from adult stem cell. Is it ethical to 'kill' a potential human life.**
- **gene therapy is very expensive, and will never be cost effective enough to merit high social priority;**

- *germ-line gene therapy experiments would involve too much scientific uncertainty and clinical risks, and the long term effects of such therapy are unknown;*
- *gene therapy would open the door to attempts at altering human traits not associated with disease, which could exacerbate problems of social discrimination;*
- *as germ-line gene therapy involves research on early embryos and effects their offspring, such research essentially creates generations of un-consenting research subjects;*
- *germ-line gene therapy would violate the rights of subsequent generations to inherit a genetic endowment that has not been intentionally modified*

[Total: 15 marks]

Question 3

A Restriction Fragment Length Polymorphism (RFLP) locus is very closely linked to an autosomal disease-causing gene.

When an individual's DNA sample is digested with *EcoRI*, it produces two bands of interest, a 7kb and an 8kb band on the Southern blot.

(a) Explain how Restriction Fragment Length Polymorphisms come about. [2]

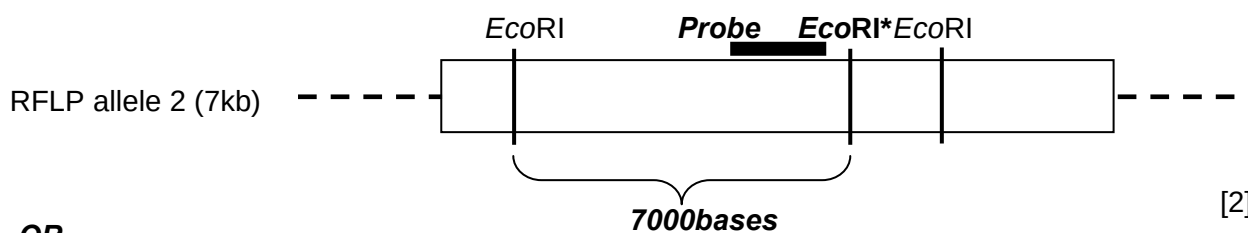
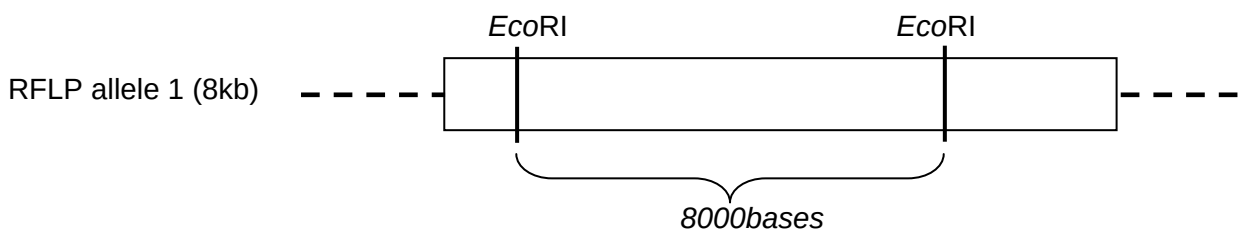
- **Difference in nucleotide sequence between chromosomes / individuals**
- **Causes restriction sites to be created at a loci / disappear**
- **When cut with a particular Restriction Enzyme**
- **Will produce Restriction fragments of differing lengths**

(b) If the RFLP allele causing the 8kb band has the *EcoRI* restriction sites as shown below, **draw on the RFLP allele that is responsible for the 7kb band**, indicating:

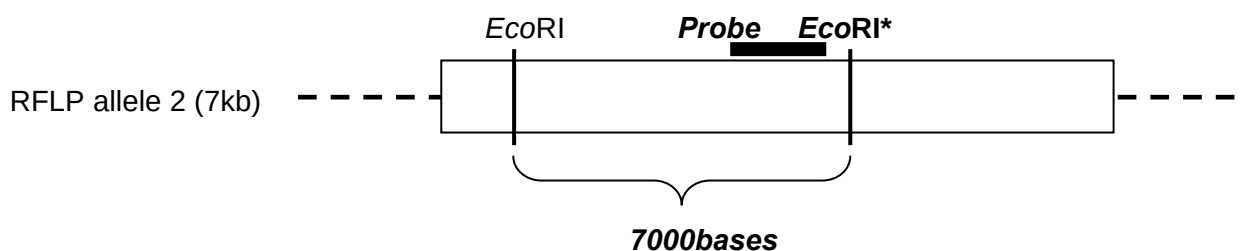
- i) the position of the polymorphic restriction site on the 7kb allele, marking the polymorphic restriction site with an asterisk (*) and
- ii) the position of the probe.

Label any important regions showing the distance between restriction sites as is demonstrated for you in the RFLP 8kb allele.

Dashes show the continuation of the rest of the chromosome. [2]



OR



Probe can be anywhere along the region of different fragment length. Probes not completely overlapping the region are also accepted.

Candidates can choose to shift one of the two *EcoRI* sites on allele 1 to be the 'polymorphic restriction site', although by definition polymorphic restriction site simply means 'present on one allele absent on the other' thus it would be better to have a third *EcoRI* site. It is possible because the question says – "portion of Southern blot" and therefore it is possible that the band for 1kb does exist but is not shown.

The figure below shows the pedigree of a family in which the same autosomal disease-causing dominant allele is transmitted through two generations. The figure also shows a corresponding portion of a Southern blot with DNA samples from each individual in the family, digested with *EcoRI* and probed with a DNA fragment that detects a RFLP.

Squares indicate males and circles indicate females, shaded squares and/or circles indicate that the individual is affected by the disease.

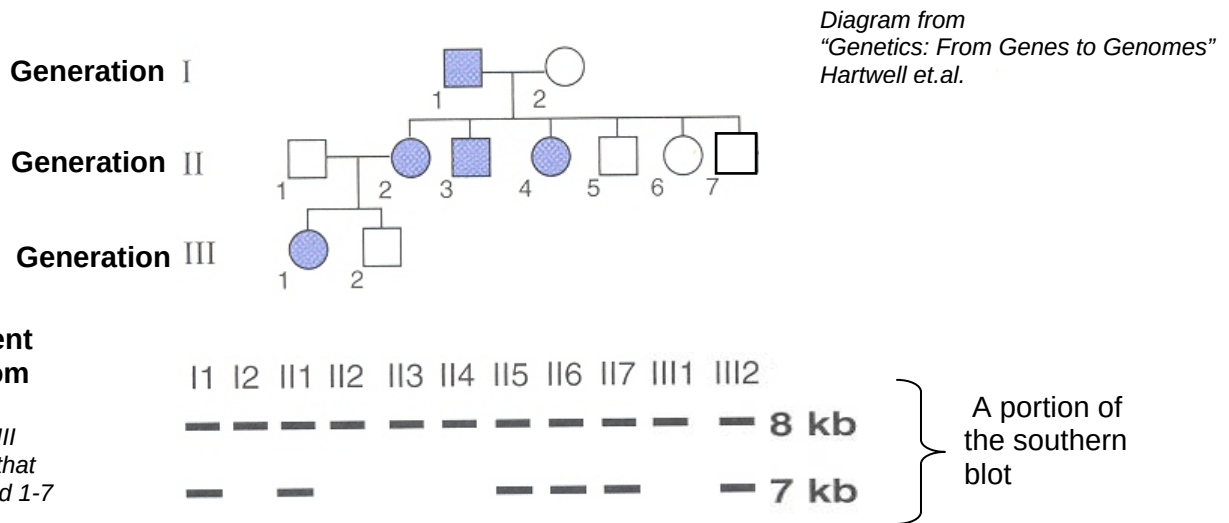


Fig. 3.1

Look at the pattern of inheritance of the disease-causing dominant allele and the RFLP 8kb and 7kb alleles in the pedigree chart. (Fig. 3.1)

- (c) i) Which allele, 8kb (allele 1) or 7kb (allele 2) is always inherited together with the disease-causing allele? [1]

8kb allele

- ii) With reference to the transmission of alleles from Generation I to Generation II, explain your answer in (c) i). [3]

- **II2, II3 and II4 are affected and are homozygous for the 8kb allele.**
- **Thus they must have received one 8kb allele from their father I1 who is affected and thus also has the disease causing allele**
- **II5, II6 and II7 must have received the 7kb allele from father I1 because the mother is homozygous for the 8kb allele. They are not affected, and thus do not have the disease causing allele.**

Answer that uses pattern of "most of those with 8kb allele also have disease" will not be accepted because I2, II5-7 also have 8kb allele but do not suffer from the disease.

(d) Outline the steps required to get from the binding of probe to the image shown in the southern blot. [2]

- **Probe that is on the rest of the nitrocellulose membrane is then washed off, leaving the probe bound only to the complementary DNA**
- **Depending on what kind of probe was used, radioactive probe – autoradiography will be used to produce the image shown, or for fluorescent probes – UV light will be shone onto the membrane to produce the image shown.**

(e) Explain how such RFLP markers have aided in genomic mapping. [3]

- **Increased number of markers besides gene markers**
- **Non-coding regions which may make up a large proportion of the genome, can be mapped**
- **Distances in terms of nucleotides can be determined from restriction fragment lengths**

(f) How is genomic mapping advantageous as compared to obtaining the sequence of all the genes in an organism? [2]

- **Relative position of genes may not be known from the sequence alone, thus genomic mapping allows for both position and sequence to be known**
- **Regulatory sequences can be studied and therefore mechanisms for the switching on / off of the genes can be better known.**

SECTION B: 20 MARKS

Question 4

- a) Explain in detail, the key stages of producing a recombinant herbicide resistant plant in large quantities. [8]
- b) State the advantages and disadvantages of cloning plants in the laboratory. [6]
- c) Describe the possible moral concerns of using animals for genetic engineering experiments. [6]

(a) Explain in detail, the key stages of producing a recombinant herbicide plant in large quantities [8]

- **leaf fragment technique**
- **cut discs from leaf**
- **disarm ti plasmid/ t-dna gene removed**
- **replaced with gene of interest**
- **transform into agrobacterium tumefaciens**
- **culture in medium with recombinant/genetically modified bacteria**
- **ref to any screening process for either bacteria or selecting plant cells**
- **DNA of Ti plasmid integrates with the DNA of host cell**
- **Grow in medium containing nutrients, pgr and agar.**
- **Varying pgr/hormones to initiate root or shoot formation**
- **Acclimatize to similar environmental conditions**
- **Subculture callus to produce many calli to grow in large quantities.**

1 mark each

(b) State the advantages and disadvantages if cloning plants in the laboratory. [6]

Advantages

- ***Rapid and mass multiplication of plants possible***
- ***Genetic makeup of plants possessing desirable phenotypes can be preserved generation after generation***
- ***Overcomes problems of heterozygotes that do not breed true***
- ***Disease-free plants in meristematic culture***
- ***Takes up little space, so can be grown intensively***
- ***Used to propagate plant species with very low seed germination rates/sterile plants***
- ***Produced out of season to meet consumer demand***
- ***Air freighted easily and cheaply, because cultures are not bulky***
- ***Standardization of growth conditions, ensuring product uniformity***
- ***Genetically engineered plants can be multiplied and mass produced relatively easily***

Disadvantages

- ***Contamination of cultures may result in high losses***
- ***Costly and tedious processes, therefore not suitable for crops with low market value***
- ***May produce many genetic offshoots which are undesirable***
- ***Limited gene pool makes them vulnerable to new disease or change in environment.***

½ mark each

(c) Describe the possible negative moral concerns of using animals for genetic engineering experiments. [6]

- ***issue of unnaturalness/playing god/altering mother nature***
- ***cross species transfer may compromise the integrity of the genetic makeup of species***
- ***may not be acceptable to various religions or beliefs***
- ***still imprecise science/elementary stages***
- ***inserted genes may show undesirable effects on long term***
- ***should be labeled to allow consumers the right to choose, if products are released in market***
- ***modified food may not be acceptable to various religions or vegans***
- ***food should be labeled to give consumers the right to choose***
- ***Research experiments induces suffering***
- ***Animals are also sentient creatures capable of feeling pain and have emotions***
- ***Unequal treatment to animals***
- ***Side effects also may cause unnecessary harm to animal/future generations***
- ***Disease models to intentionally induce suffering in animals***

- ***Sanctity of life argument***
- ***All forms of life are intrinsically sacred and animals are deprived of the right to live in these experiments***

- ***Some experiments does not increase the welfare of human/or animal/or increase the greater good for mankind***
- ***Not for the urgent needs of mankind***

- ***A balance must be struck between the well-being and treatment of these animals and the desires of mankind***
- ***Slippery slope argument***
- ***Controversial experiments like Cloning may be done on humans in the future.***
- ***Continued poverty in farmers of poor country***
- ***If gm suppliers could make farms from developing nations have heavy reliance on them for stock supplies***
- ***Patenting of animals reduces them only to the level of objects/commodities***

[1/2] mark per point] max 1 mark for any six forms of arguments.

Explain in detail, the key stages of producing a recombinant herbicide plant in large quantities [8]

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State the advantages and disadvantages of cloning plants in the lab [6]

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[1/2 mark per point] max 1 mark for any six forms of arguments.

END OF PAPER 3